All-pair arithmetic on the growing smooth family

For every t, Lean gives the actual reduced pairs identity and $(X^d,\pm1)$, $1\le d\le g$, a cyclic group law over F_3 for $f=X^n+1$, $g=12t+2$, $n=24t+5$. Every sum equals an independently defined unique normalized Cantor output. The actual generator (X,1) has exact order n and generates every member. The earlier polynomial-defined cycle step equals addition by that generator. Genus and order grow without bound. [pcc:unverified]

All group members are smooth, fraction 1. The full finite set of valid reduced pairs has smooth fraction strictly smaller than 1, witnessed by (X^2+X+2,X). This verifies the group law on the family. It leaves the full ambient group and geometric Picard correspondence open. [pcc:unverified]

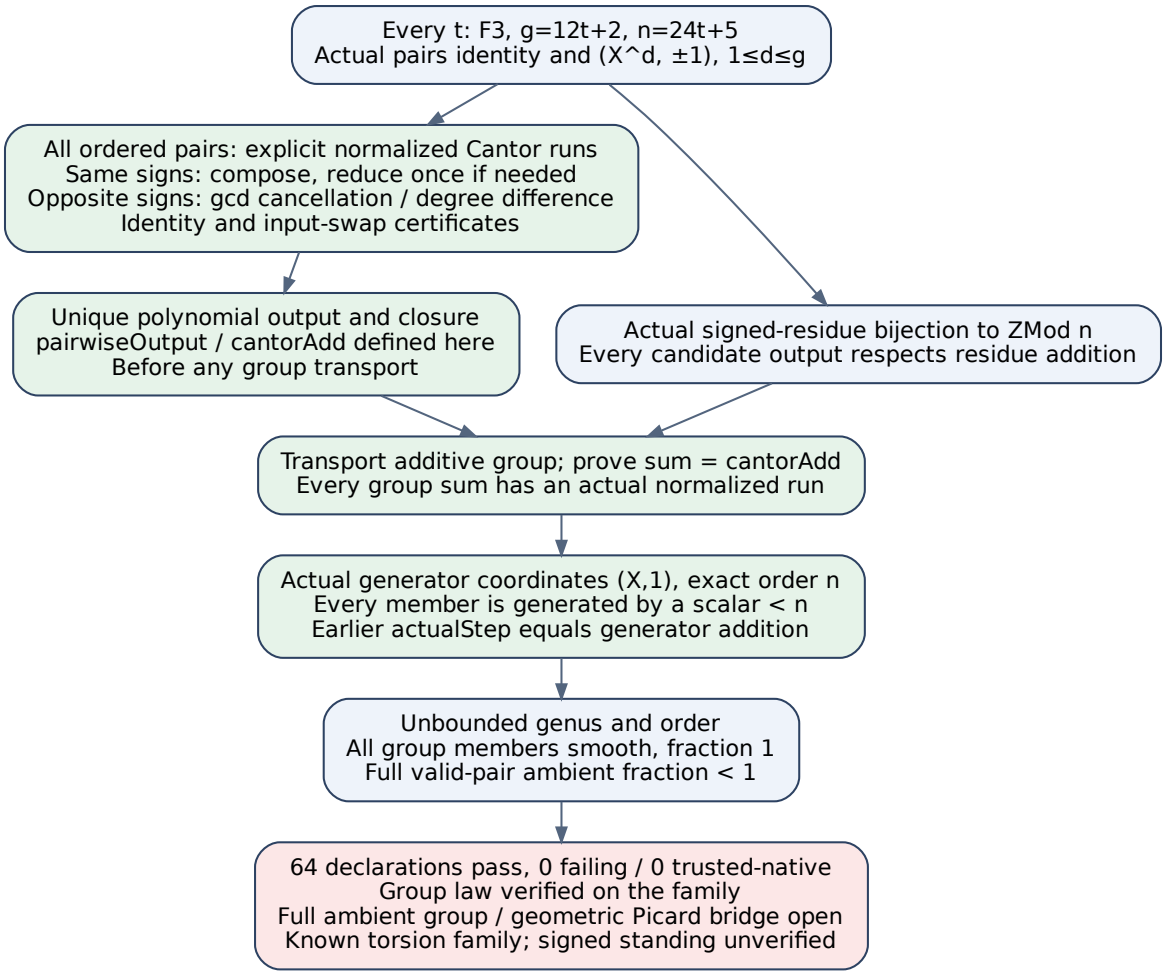

Figure 1: Actual arithmetic precedes and agrees with group transport. [pcc:unverified]

Certificates, source scope and remaining bridge

Same-sign inputs $(X^a,s),(X^b,s)$, $s=\pm1$, have monic gcd 1. Their composition has degree a+b. If it exceeds g, one normalized reduction gives $(X^{n-a-b},-s)$. Opposite-sign inputs have gcd $X^{\min(a,b)}$: equal degrees cancel to identity, and unequal degrees leave the larger sign and degree $|a-b|$. Full polynomial gcd/Bézout, composition, bounded remainder and reduction equations are certified for every ordered pair; identity and swap cases are included. [pcc:unverified]

The signed residue map is a bijection to ZMod n. Every candidate output respects residue addition. Output uniqueness holds for explicitly selected Bézout coefficients, so pairwiseOutput and cantorAdd are defined from the actual normalized equations before group transport. The transported sum is then proved equal to cantorAdd. The actual generator has order n; every member is j times it for some $j<n$. A coefficient and actual-point equality proof binds the frozen parent's actualStep to this same generator addition. [pcc:unverified]

Module	Exact checked obligation
AllPairRuns	General same/opposite-sign, cancellation, identity and swap polynomial runs
IndexArithmetic	Signed-residue bijection; every candidate respects addition
PairwiseOutput	All pairwise runs, unique output and closure before group structure
VerifiedFamilyGroup	Sum equals actual normalized arithmetic; generator order and cyclicity
GeneratorAgreement	Earlier actualStep equals generator addition; combined unbounded-family certificate

The whole project audit passes 64 declarations, zero failures and zero trusted-native exceptions. Lean 4.35.0 / Mathlib 56a00585, standard axioms only: propext, Classical.choice and Quot.sound. No holes, custom axioms, native_decide, linter waivers or added elaboration settings. All 22 attempt/source files and five gate files have byte-equal archive checks. Issued parent formal sources remain unchanged. Exact source commit, packet, local verdict and signed standing appear in verification.json; signed standing remains unverified. [pcc:unverified]

The exact primary read receipts are reused: Stoll 2014, Lemmas 4.16–4.17, Theorem 4.18 and Example 4.19, printed pages 23–26; Jędrzejak 2016, relevant introduction/theorem, model, good-reduction and printed-page-111 torsion proof. No new full-paper read is claimed. Cantor's original remains unread after ordinary AMS HTTP403. Source-review.json pins the prior URLs, hashes and read scopes. No novelty claim. [pcc:unverified]

The full ambient reduced-pair group law and its geometric Picard/Riemann–Roch identification remain separate formal obligations. No unproved ambient embedding is assumed. The orders 24t+5 are not all prime. No refutation of fixed-genus $q\to\infty$ or typical/large-prime subgroup asymptotics is claimed. [pcc:unverified]

The existing IDC1hb7102c15384f2648 is copied byte-identically for its base-quintic scalar identity only. It does not certify this field, variable exponent, group law, period, counting or geometry; the new arithmetic rests on Lean. No new native experiment, sampling, timing or paid compute. KB/Conductor unavailable; bounded primary/repository/Mathlib fallback. Proof map and pins: README.md, sources.json, verification.json. [pcc:unverified]